EXISTENCE OF A CLOSED-IMMERSION SOLUTION FROM BOUNDARY DATA

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5516-5527 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 5529-5610, the complete author proof (80 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_artin.json (pb=3401, stmt 5516-5527, proof 5529-5610).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-closed-immersion' also exists in stacks-descent, stacks-sites — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05ER: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GD6: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AQT: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH4: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH6: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH5: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 073K: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AS: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AZ: sources/stacks-artin/pinned-original-source.tex, source lines 5610–5610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 assume given a closed subset $Z \subset S$ such that*

(1) *the inverse image of Z in X' is T',*
(2) *$U' \to S \setminus Z$ is a closed immersion,*
(3) *$W \to S_{/Z}$ is a closed immersion.*

Then there exists a solution $(f : X' \to X, T, a)$ and moreover $X \to S$ is a closed immersion.

Proof. Suppose we have a closed subscheme $X \subset S$ such that $X \cap (S \setminus Z) = U'$ and $X_{/Z} = W$. Then X represents the functor F (some details omitted) and hence is a solution. To find X is clearly a local question on S. In this way we reduce to the case discussed in the next paragraph.

Assume $S = \mathrm{Spec}(A)$ is affine. Let $I \subset A$ be the radical ideal cutting out Z. Write $I = (f_1, \ldots, f_r)$. By assumption we are given

(1) the closed immersion $U' \to S \setminus Z$ determines ideals $J_i \subset A[1/f_i]$ such that J_i and J_j generate the same ideal in $A[1/f_i f_j]$,
(2) the closed immersion $W \to S_{/Z}$ is the map $\mathrm{Spf}(A^\wedge/J') \to \mathrm{Spf}(A^\wedge)$ for some ideal $J' \subset A^\wedge$ in the I-adic completion $A^\wedge$ of A.

To finish the proof we need to find an ideal $J \subset A$ such that $J_i = J[1/f_i]$ and $J' = JA^\wedge$. By More on Algebra, Proposition 05ER it suffices to show that J_i and J' generate the same ideal in $A^\wedge[1/f_i]$ for all i.

Recall that $A' = H^0(X', \mathcal{O})$ is a finite A-algebra whose formation commutes with flat base change (Cohomology of Spaces, Lemmas 08AS and 073K). Denote $J'' =$

$\mathrm{Ker}(A \to A')$[1]. We have $J_i = J''A[1/f_i]$ as follows from base change to the spectrum of $A[1/f_i]$. Observe that we have a commutative diagram

$$\begin{array}{ccccc} X' & \longleftarrow & X'_{/T'} \times_{S_{/z}} \mathrm{Spf}(A^\wedge) & = & X'_{/T'} \times_W \mathrm{Spf}(A^\wedge/J') \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longleftarrow & \mathrm{Spf}(A^\wedge) & \longleftarrow & \mathrm{Spf}(A^\wedge/J') \end{array}$$

The middle vertical arrow is the completion of the left vertical arrow along the obvious closed subsets. By the theorem on formal functions we have

$$(A')^\wedge = \Gamma(X' \times_S \mathrm{Spec}(A^\wedge), \mathcal{O}) = \lim H^0(X' \times_S \mathrm{Spec}(A/I^n), \mathcal{O})$$

See Cohomology of Spaces, Theorem 08AZ. From the diagram we conclude that J' maps to zero in $(A')^\wedge$. Hence $J' \subset J''A^\wedge$. Consider the arrows

$$X'_{/T'} \to \mathrm{Spf}(A^\wedge/J''A^\wedge) \to \mathrm{Spf}(A^\wedge/J') = W$$

We know the composition g is a formal modification (in particular rig-étale and rig-surjective) and the second arrow is a closed immersion (in particular an adic monomorphism). Hence $X'_{/T'} \to \mathrm{Spf}(A^\wedge/J''A^\wedge)$ is rig-surjective and rig-étale, see Algebraization of Formal Spaces, Lemmas 0GH5 and 0GH4. Applying Algebraization of Formal Spaces, Lemmas 0GH6 and 0AQT we conclude that $\mathrm{Spf}(A^\wedge/J''A^\wedge) \to W$ is rig-étale and rig-surjective. By Algebraization of Formal Spaces, Lemma 0GD6 we conclude that $I^nJ''A^\wedge \subset J'$ for some $n > 0$. It follows that $J''A^\wedge[1/f_i] = J'A^\wedge[1/f_i]$ and we deduce $J_iA^\wedge[1/f_i] = J'A^\wedge[1/f_i]$ for all i as desired. □

[1]Contrary to what the reader may expect, the ideals J and J'' won't agree in general.